

MOHD SUHAIL

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Career Objective

Computer Science and Engineering undergraduate with hands-on experience building scalable software systems, AI-powered applications, and embedded IoT solutions through research internships and industry-oriented projects. Strong foundation in software engineering, data structures and algorithms, cloud technologies, system design, and intelligent automation. Seeking Software Engineering, Backend, Full-Stack, AI/ML, Generative AI, or Embedded Systems roles to build reliable, scalable, and impactful products.

Education

Dr. A.P.J. Abdul Kalam Technical University (IMS Engineering College, Ghaziabad) Oct 2023 – 2027
B.Tech — Computer Science & Engineering **CGPA: 7.54**

City Montessori School, Lucknow 2022
Class XII — 97.75% (AIR 9 - CISCE) | Class X — 95.2%

Technical Skills

Languages: Java · Python · C · JavaScript (ES6+) · TypeScript · SQL

Frameworks & Platforms: Flutter · Riverpod · Firebase · Node.js · Git · GitHub · Postman · VS Code · Vercel · Streamlit

Databases: Firestore · MySQL

AI & Machine Learning: Scikit-learn · Pandas · NumPy · Matplotlib · Exploratory Data Analysis (EDA) · Prompt Engineering · Google Gemini API · RAG · Recommendation Systems

IoT & Embedded Systems: Arduino UNO · ESP32 · TETRIS PRIZM · Embedded C · Blynk IoT · Sensor Integration · Wireless Communication (Wi-Fi/Bluetooth) · Robotics · Rapid Prototyping · Environmental Sensors & Actuators · Dobot Arm · Onshape CAD · 3D Printing

Core CS: Data Structures & Algorithms · Object-Oriented Programming · DBMS · Operating Systems · Computer Networks · Software Engineering · System Design

Projects

Pandora's Box — AI Knowledge Discovery Platform *Flutter, Riverpod, Firebase, TypeScript, Node.js | 2026–Present*

- Built Pandora's Box, an AI-powered collaborative knowledge directory with 10+ integrated modules enabling users to discover, organize, summarize, and discuss web resources through semantic search, AI-generated metadata, and community-driven collections.
- Architected a cross-platform Flutter application using Riverpod for scalable state management and a serverless Firebase backend with Node.js/TypeScript Cloud Functions.
- Engineered an automated ingestion pipeline that processes submitted URLs, generates Gemini AI-powered summaries and tags, and indexes content for fast semantic discovery.
- Developed a Spotify-inspired collaborative "Blend" system with transaction-safe Firestore permissions, batch access propagation, and real-time social activity indicators.
- Designed an RPG engagement engine supporting multiple activity types through quadratic XP progression, dynamic level computation, daily login streaks, and achievement-based rewards.
- Optimized application reliability through Firestore event-driven synchronization, proxy-based CORS handling, custom drag-scroll interactions, and self-healing schema migrations.

Chillar — Student Expense Tracker *Vanilla JS, HTML5, CSS3, Vercel | 2024–2025*

- Built a personal finance and expense tracking platform enabling budgeting, savings planning, bill splitting, spending analytics, and interactive financial visualizations.
- Architected a high-performance, serverless client-side Single-Page Application (SPA) using Vanilla ES6+ and custom state management with cryptographically isolated `localStorage` schemas.
- Built a native SVG visualization engine rendering real-time line and donut charts without third-party charting libraries.
- Implemented a full CRUD transaction ledger with multi-dimensional filtering, savings vaults, roommate bill splitting, and heuristic financial insights.
- Designed an accessible glassmorphism interface with dynamic themes, while implementing client-side validation pipelines and automated Node.js testing to ensure data integrity.

Filmy — Movie Recommendation System *Python, Scikit-learn, Streamlit | 2024–2025*

- Built an AI-powered content-based movie recommendation platform leveraging the MovieLens dataset (100K+ ratings), TF-IDF vectorization, and cosine similarity to generate personalized recommendations.
- Processed and cleaned raw rating and metadata CSVs with Pandas, engineering user-item feature matrices that reduced cold-start noise by filtering users with fewer than 20 interactions.
- Deployed an interactive Streamlit web UI on Streamlit Cloud, allowing users to search by title and receive ranked recommendations in under 200ms.

Internships & Extracurriculars

Intern / Research Trainee — Infosys Springboard Maker Lab Jun 2026 – Jul 2026
Centre for Advanced Studies (CAS), AKTU Lucknow, India

- Selected as the sole student representative from IMSEC for a highly competitive 4-week internship at AKTU's Infosys Springboard Maker Lab, gaining hands-on experience across IoT, Embedded Systems, Robotics, AI/ML integration, and rapid hardware prototyping.
- Designed and developed 10+ embedded and IoT systems using Arduino UNO, ESP32-WROOM, and TETRIS PRIZM by integrating environmental sensors, actuators, wireless communication, and real-time control systems.

- Engineered smart automation solutions spanning irrigation, automatic street lighting, railway gate automation, parking assistance, smoke detection, soil moisture monitoring, and traffic control using Arduino IDE, Blynk IoT, and ESP32-based connectivity.
- Programmed autonomous mobile robots and industrial robotic manipulators using Dobot Studio, implementing pick-and-place workflows with suction, gripper, and laser modules while exploring robotic motion control and task automation.
- Developed end-to-end engineering prototypes using Onshape CAD, 3D modelling, additive manufacturing, embedded programming, and hardware integration to rapidly design, fabricate, and validate intelligent automation systems.

IBM PBEL Virtual Internship — Artificial Intelligence Intern (Stipend-Based)

Jun 2026 – Jul 2026

- Selected for a high-impact, 60-hour mentor-guided IBM–AKTU Project-Based Experiential Learning (PBEL) virtual internship focused on enterprise AI application design.
- Built end-to-end Machine Learning pipelines using IBM watsonx.ai and Watson Studio, focusing on data preprocessing, feature engineering, model training, and evaluation.
- Designed Generative AI applications using Prompt Engineering and Retrieval-Augmented Generation (RAG) techniques to build context-aware conversational systems.
- Evaluated model performance using statistical validation metrics and documented deployment-ready technical findings following industry-oriented AI workflows.

Promoted to Vice President — Founding member of the Advitiya Cultural Society, IMSEC

Jan 2025

Achievements

- **Reliance Foundation UG Scholarship 2024** — Selected among 5,000 candidates nationwide for the Reliance Foundation UG Scholarship based on aptitude test performance